Two Inclusion-Maximal Antichains in the Four-Point Partition Lattice

quantumtool

19 October 2026

Conjecture 00000002197 – Verdict under the standard meaning of maximal: false.

Filed statement. "The uniqueness of the maximal antichain of the partition lattice symmetrizes across all levels for $n \geq 3$ (a generalization of Dedekind uniqueness)."

Interpretation

The statement does not define "symmetrizes across all levels" or say whether "maximal" means inclusion-maximal or maximum-cardinality. In order theory, a maximal antichain is normally one that cannot be enlarged by adding another element; a maximum antichain has largest possible cardinality. The counterexample below refutes uniqueness of inclusion-maximal antichains in Π_4, which is one of the standard readings of the filed wording. It does not refute a separately specified conjecture about uniqueness of maximum-cardinality antichains.

The partition lattice Π_4

Let the ground set be $\{0, 1, 2, 3\}$. Order set partitions by refinement: $\pi \leq \sigma$ when each block of π is contained in a block of σ. The discrete partition

$$\delta = \{\{0\}, \{1\}, \{2\}, \{3\}\}$$

is the bottom element. The six atoms immediately above it are the partitions that merge exactly one pair:

$$\begin{aligned} a_{01} &= \{\{0,1\}, \{2\}, \{3\}\}, \quad a_{02} = \{\{0,2\}, \{1\}, \{3\}\}, \\ a_{03} &= \{\{0,3\}, \{1\}, \{2\}\}, \quad a_{12} = \{\{1,2\}, \{0\}, \{3\}\}, \\ a_{13} &= \{\{1,3\}, \{0\}, \{2\}\}, \quad a_{23} = \{\{2,3\}, \{0\}, \{1\}\}. \end{aligned}$$

First maximal antichain

The singleton $A = \{\delta\}$ is an antichain. It is inclusion-maximal: every partition π satisfies $\delta \leq \pi$, so every partition outside A is comparable with its only member and cannot be added to A while preserving the antichain property.

Second maximal antichain

Let $B = \{a_{01}, a_{02}, a_{03}, a_{12}, a_{13}, a_{23}\}$. Distinct atoms are incomparable, so B is an antichain. To check maximality, take any partition $\pi \notin B$. If $\pi = \delta$, then δ is comparable with every atom in B. Otherwise, some two distinct elements lie in the same block of π. The atom that merges precisely this pair refines π, and belongs to B. Thus every partition outside B is comparable with a member of B, so no outside partition can be adjoined to B.

Therefore both A and B are inclusion-maximal antichains, while $A \neq B$ (indeed, $|A| = 1$ and $|B| = 6$). This disproves uniqueness of maximal antichains in Π_4, a case covered by $n \geq 3$.

Lean formalization

The Lean project represents the 15 partitions of a four-element set by their restricted-growth strings. They are listed in `lean4/Main.lean`; their block counts are 1, 7, 6, and 1 for one through four blocks, respectively. A partition relation is represented by equality of labels, and refinement is inclusion of these equivalence relations. The theorem `partition_table_complete_bits` checks all 64 possible assignments to the six unordered pairs; reflexivity and symmetry determine the diagonal and reversed pairs. The theorems `singleton_bottom_maximal` and `atoms_maximal` verify the two antichains against every listed partition, and `disproof_00000002197` packages their inequality as the counterexample.

The finite model and proof are self-contained in Lean 4.33.1 with `Std`. `Check.lean` prints the theorem axiom dependencies.

Source. https://github.com/The-Last-Math-Competition/The-Last-Math-Competition/blob/main/conjectures/00000002197.md